

Highlights

- Inverse-propensity and doubly robust graph propagation are biased beyond one hop.
- DRUP removes the repeated-walk bias exactly for any odd number of hops.
- Under slate dependence its bias is second order in the imputation error.
- Training-free DR propagation beats tuned trained models on KuaiRec.
- Thinning audits confirm exposure invariance on Coat and KuaiRec only.